

NUMBER THEORY

Subtitle

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13 April 2026

OUTLINE

1. NUMBER THEORY

Division

Let a and b be integers with $a \neq 0$.

a divides b if $\exists c$ such that c is an integer and $b = ac$.

- When a divides b (notation $a \mid b$):
 - a is a factor of b .
 - b is a multiple of a .

a does not divide b is written $a \nmid b$.

e.g. $3 \mid 12$, $4 \mid 16$, $3 \nmid 5$

Theorem

Let a , b and c be integers, where $a \neq 0$.

- (i) If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$,
- (ii) If $a \mid b$, then $a \mid bc$ for all integers c ,
- (iii) If $a \mid b$ and $b \mid c$, then $a \mid c$.

For example,

- (i) If $2 \mid 4$ and $2 \mid 6$, then $2 \mid (4 + 6) \Rightarrow 2 \mid 10$,
- (ii) If $2 \mid 8$, then $2 \mid 8 * 4c$, $2 \mid 8 * 7$
- (iii) If $3 \mid 6$ and $6 \mid 12$, then $3 \mid 12$.

Theorem

Let a , b and c be integers, where $a \neq 0$.

- (i) If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$,

Proof:

Spse $a \mid b$ and $b \mid c$

Therefore, exists two integers s and t such that

$b = as$ and $c = at$ def of divisibility

$$b + c = as + at = a(s + t)$$

$b + c = a(s + t)$, that is $a \mid (b + c)$

- (ii) If $a \mid b$, then $a \mid bc$ for all integers c ,

Proof:

Spse $a \mid b$ then $b = at$ for some $t \in \mathbb{Z}$

$$bc = (at)c = a(tc)$$

Therefore, $a \mid bc$ by def of divisibility.

(iii) If $a \mid b$ and $b \mid c$, then $a \mid c$.

Proof:

Spse $a \mid b$ then $b = at$ for some $t \in \mathbb{Z}$

Spse $b \mid c$ then $c = bs$ for some $s \in \mathbb{Z}$

$$c = (at)s = a(ts)$$

Therefore, $a \mid c$ by def of divisibility.

Corollary

If a , b and c are integers, where $a \neq 0$, such that $a \mid b$ and $a \mid c$, then $a \mid mb + nc$ whenever n and m are integers.

The Division Algorithm

Theorem:

Let a be an integer and d a positive integer. Then there are unique integers q and r , with $0 \leq r < d$, such that:

$$a = dq + r.$$

d is the divisor, a is the dividend

q is the quotient; $q = a \text{ div } d$

r is the remainder; $r = a \text{ mod } d$

Example

What are the quotient and remainder when

(i) 101 is divided by 11?

Solution: We have $101 = 11 \cdot 9 + 2$. Hence, the quotient when 101 is divided by 11 is $9 = 101 \text{ div } 11$, and the remainder is $2 = 101 \text{ mod } 11$.

(ii) -11 is divided by 3?

Solution: We have $-11 = 3(-4) + 1$. Hence, the quotient when -11 is divided by 3 is $-4 = -11 \text{ div } 3$, and the remainder is $1 = -11 \text{ mod } 3$.

Note that the remainder cannot be negative.

Modular Arithmetic

Def - Congruence:

Let a and b be integers and m a positive integer. a is said to be congruent to b modulo m if m divides $a - b$. This means that:

- $a \bmod m = b \bmod m$, i.e. a and b have the same remainder when divided by m .
- Above is written as: $a \equiv b \pmod{m}$ and read as: a is congruent to b modulo m .
- If a is not congruent to b modulo m , we write it as: $a \not\equiv b \pmod{m}$.

Example:

Determine whether:

- 17 is congruent to 5 modulo 6, (i.e. $17 \equiv 5 \pmod{6}$),
- 24 and 14 are congruent modulo 6? (Is 24 congruent to 14 modulo 6?).

Solution:

- Yes, $17 \equiv 5 \pmod{6}$, $6 \mid 17 - 5$, $6 \mid 12$
- No, $24 \not\equiv 14 \pmod{6}$, $6 \mid 24 - 14$, $6 \nmid 10$

Theorem:

Let a and b be integers, and let m be a positive integer. Then $a \equiv b \pmod{m}$ if and only if

$$a \bmod m = b \bmod m.$$

Example:

$$17 \equiv 5 \pmod{6}$$

$$\Rightarrow 17 \bmod 6 = 5 \bmod 6$$

Theorem:

Let m be a positive integer. The integers a and b are congruent modulo m if and only if there is an integer k such that $a = b + km$.

Proof:

Let $a \equiv b \pmod{m}$. Then $m \mid (a - b)$ by def of congruence.

Then $a - b = km$ for some $k \in \mathbb{Z}$ by def of divisibility.

By addition of b , $a = b + km$

Assume $a = b + km$, then $a - b = km$ by subtraction.

Then, $m \mid (a - b)$, so $a \equiv b \pmod{m}$.

Example:

$$17 \equiv 5 \pmod{6}$$

$$\Rightarrow 17 = 5 + 6.k$$

$$\Rightarrow 17 = 5 + 6.2$$

Theorem:

Let m be a positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then:

- $a + c \equiv b + d \pmod{m}$, and
- $ac \equiv bd \pmod{m}$

Proof:

Assume $a \equiv b \pmod{m}$. Then, $a = b + km$ for some $k \in \mathbb{Z}$

Assume also, $c \equiv d \pmod{m}$. Then, $c = d + lm$ for some $l \in \mathbb{Z}$

$$a + c = (b + km)(d + lm) = b + d + (km + lm) = (b + d) + m(k + l)$$

$$(a + c) = (b + d) + m(k + l), \text{ so } (a + c) - (b + d) = m(k + l), \text{ so}$$

$$m \mid (a + c) - (b + d), \text{ so } a + c \equiv b + d \pmod{m}.$$

$$ac = (b + km)(d + lm) = bd + blm + kdm + klm^2 = bd + m(bl + kd + klm)$$

$$\text{So, } ac - bd = m(bl + kd + klm), \text{ so } m \mid ac - bd, \text{ so } ac \equiv bd \pmod{m}$$

Example:

Because $7 \equiv 2 \pmod{5}$ and $11 \equiv 1 \pmod{5}$, it follows from the Theorem that

$$18 = 7 + 11 \equiv 2 + 1 = 3 \pmod{5}$$

and that,

$$77 = 7.11 \equiv 2.1 \pmod{5}$$

Example:

Because $7 \equiv 2 \pmod{5}$ and $11 \equiv 1 \pmod{5}$, it follows from the Theorem that $18 = 7 + 11 \equiv 2 + 1 = 3 \pmod{5}$

and that,

$$77 = 7.11 \equiv 2.1 \pmod{5}$$

Example: Find the integer a such that

a. $a \equiv 8 \pmod{5}$, and $0 \leq a \leq 5$

Sol:

8 divided by 5 = 1 remainder 3, so $8 \equiv 3 \pmod{5}$

Thus, $a \equiv 3 \pmod{5}$. $\therefore a = 3$.

b. $a \equiv -15 \pmod{27}$, and $0 \leq a \leq 27$

Sol:

Add 27 to -15 to get a positive remainder: $-15 + 27 = 12$. Hence $-15 \equiv 12 \pmod{27}$

So $a \equiv 12 \pmod{27}$. Within $[0, 27]$, the only solution is $a = 12$.

c. $a \equiv 99 \pmod{41}$, and $100 \leq a \leq 140$

Sol:

Divide 99 by 41: $41 \times 2 = 82$, remainder $99 - 82 = 17$. Thus $99 \equiv 17 \pmod{41}$

So $a = 17 + 41k$ for some integer k

Finding k such that $100 \leq 17 + 41k \leq 140$:

For $k = 2$: $17 + 41 \cdot 2 = 99$ (too low)

For $k = 3$: $17 + 41 \cdot 3 = 140$ (within range)

For $k = 4$: $17 + 41 \cdot 4 = 181$ (too high)

Thus, $k = 3$

Example:

Suppose that a and b are integers, $a \equiv 4 \pmod{13}$, and $b \equiv 9 \pmod{13}$. Find the integers c with $0 \leq c \leq 12$ such that:

- a. $c \equiv 9a \pmod{13}$
- b. $c \equiv 11b \pmod{13}$
- c. $c \equiv a + b \pmod{13}$

Sol:

- a. $c \equiv 9a \pmod{13}$
 $c \equiv 9(4) \pmod{13}$
 $c \equiv 36 \pmod{13}$
 $10 \equiv 36 \pmod{13}$ because $36 = 13 \cdot 2 + 10$
 $c = 10$

Example (cont):

b. $c \equiv 11b \pmod{13}$

$$c \equiv 11(9) \pmod{13}$$

$$c \equiv 99 \pmod{13}$$

$$8 \equiv 99 \pmod{13} \text{ because } 99 = 13 \cdot 7 + 8$$

$$c = 8$$

c. $c \equiv a + b \pmod{13}$

$$c \equiv 4 + 9 \pmod{13}$$

$$c \equiv 13 \pmod{13}$$

$$0 \equiv 13 \pmod{13} \text{ because } 13 = 13 \cdot 1 + 0$$

$$c = 0$$

Show that if n is an integer, then $n^2 \equiv 0 \pmod{4}$ or $n^2 \equiv 1 \pmod{4}$.

Sol:

Proof

Let n be an integer. Then n can either be even or odd.

Case 1: Suppose n is even.

Then by definition of even, $n = 2k$ for which $k \in \mathbb{Z}$.

Also, $n^2 = 4k^2$. So, since $n^2 = 4k^2$, $n^2 \equiv 4k^2 \pmod{4}$. In addition, $4k^2$ is a multiple of 4, so $4k^2 \equiv 0 \pmod{4}$. If $n^2 \equiv 4k^2 \pmod{4}$, and $4k^2 \equiv 0 \pmod{4}$, then $n^2 \equiv 0 \pmod{4}$ when n is even.

Case 2: Suppose n is odd.

Then by definition of odd, $n = 2k + 1$ for some integer k .

And, $n^2 = (2k + 1)^2 = 4k^2 + 4k + 1$

So, by combining the following congruences,

$$n^2 \equiv 4k^2 + 4k + 1 \pmod{4}$$

Then, $4k^2 \equiv 0 \pmod{4}$, and $4k \equiv 0 \pmod{4}$

We conclude that $n^2 \equiv 1 \pmod{4}$ when n is odd.

Thus, for any integer n , $n^2 \equiv 0$ or $1 \pmod{4}$.

Practice: Show that for some positive odd integer n , n^2 is congruent to 1 mod 8, i.e. $n^2 \equiv 1 \pmod{8}$.

Applications:

Hashing Functions

Generating pseudorandom numbers

Cryptography

Cryptology has many applications in CS, especially when messages are sent over the network.

- Credit card payments
- Internet and Cellphone banking
- ATM transactions
- E-commerce applications such as buying goods online
- etc.

Hashing Functions

$h(k) = k \pmod{m}$, we need only compute the remainder when k is divided by m .

Example:

Find the memory locations assigned by the hashing function $h(k) = k \pmod{111}$ to the records of customers with US Social Security numbers 064212848 and 037149212.

Solution: The record of the customer with Social Security number 064212848 is assigned to memory location 14, because

$$h(064212848) = 064212848 \bmod 111 = 14.$$

Similarly, because $h(037149212) = 037149212 \bmod 111 = 65$,

the record of the customer with Social Security number 037149212 is assigned to memory location 65.

Pseudorandom Numbers

Linear congruential method: procedure for generating pseudorandom numbers. It uses four (4) integers:

- Modulus, (m)
- Multiplier, (a) such that $2 \leq a < m$.
- Increment, (c) such that $0 \leq c < m$.
- Seed, (x_0) such that $0 \leq x_0 < m$.

This generates a sequence of pseudo random numbers $\{x_n\}$, with $0 \leq x_n < m$ for all n by using the following congruence: $x_{n+1} = a.x_n + c \pmod{m}$.

Example:

What sequence of pseudorandom numbers is generated using $x_{n+1} = (4x_n + 1) \pmod{7}$ with seed $x_0 = 3$?

Sol:

In this case, the seed $x_0 = 3$, modulus $m = 7$, increment $c = 1$, and multiplier $a = 4$

We compute the terms of this sequence by successively using the recursively defined function $x_{n+1} = (4x_n + 1) \pmod{7}$, beginning by inserting the seed $x_0 = 3$ to find x_1 .

$$x_0 = 3$$

$$x_1 = (4(3) + 1) \pmod{7} = 13 \pmod{7} = 6$$

$$x_2 = (4(6) + 1) \pmod{7} = 25 \pmod{7} = 4$$

$$x_3 = (4(4) + 1) \pmod{7} = 17 \pmod{7} = 3$$

$$x_4 = (4(3) + 1) \pmod{7} = 13 \pmod{7} = 6$$

Because $x_3 = x_0$ and because each term depends only on the previous term, we see that the sequence 3, 6, 4, 3, ... is generated.

The End